CSEN 123 Mechatronics — Dylan Thornley

HW 2

1.

ppm	out Amps
100	7m
200	16m
300	36m
400	60m
500	96m

a. $300 < 330 < 400$

$$\frac{60m - 36m}{400-300} = \frac{x - 36m}{330-300}$$

$x = \boxed{43.2mA}$

b. $16 < 20 < 36$

$$\frac{(36-16)m}{300-200} = \frac{(20-16)m}{x-200}$$

$x = \boxed{220 \text{ ppm}}$

2.

$5/80 = 62.5mA$ ← max current w/ no LED

a. $V_f = 3.4$ (typical)

$$\frac{5-3.4}{80} = 20mA$$

also perfectly $= I_f$

$\boxed{I_f = 20mA}$

b. $V_f = 1.8$ (typical)

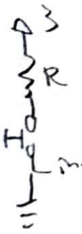
$5 - 1.8 = 3.2$

$\frac{3.2}{80} = 40mA$, exceeds 10mA max

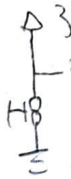
typical says 2mA, but calculated is 40. I believe the actual current will be closer to 40mA and pop the red LED

3.

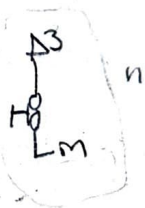
blows up X



always 0 X

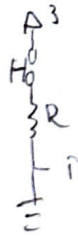


blows up X



no gnd, so 0 will read as hi-z. X

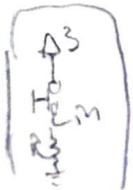
blows up X



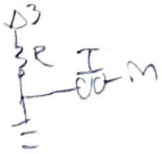
always 0 X

floating gnd. no voltage drop X

3V when not pressed, 0V when pressed X



standard switch circuit ✓



always 0 or Hi-z X